

Set A

Kishoreganj University
Department of Computer Science and Engineering
3rd year 1st semester final examination 2024

Course Title: Compiler Design Lab

Program: B.Sc. (Engg.)

Course Code: CSE 3104

Full Marks: 40

Session: 2021-22

Total Time: 1.5 Hours

Answer the following questions.

Marks

1. Your university is developing a documentation extraction tool that analyzes C programs. One feature must automatically detect and print all comments (both single-line and multi-line) from a given .c file.

20

Task:

Write a C program that reads a .c file and prints all comments found in the file.

Your program must detect:

- Single-line comments starting with //
- Multi-line comments starting with /* and ending with */
- Ignore comment markers inside strings.

2. Write a C program to construct the LL(1) parsing table for the following grammar.

20

$E \rightarrow TE'$

$E' \rightarrow +TE' \mid \epsilon$

$T \rightarrow FT'$

$T' \rightarrow * FT' \mid \epsilon$

$F \rightarrow (E) \mid id$

Or

Write a C program to remove left recursion from the following grammar.

$E \rightarrow E + T \mid T$

$T \rightarrow T * F \mid F$

$F \rightarrow (E) \mid id$